

MARTINI 3 COARSE-GRAINED FORCEFIELD FOR NON-CANONICAL AMINO ACIDS.

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ABSTRACT

Peptides and proteins represent increasingly important targets for therapeutic development due to their high specificity and favorable toxicity profiles. However, their application is often limited by challenges such as poor membrane permeability, low metabolic stability, and limited aqueous solubility. Side-chain and backbone modifications, including the incorporation of non-canonical amino acids (ncAAs), offer viable routes to overcome these limitations. Yet, the vast chemical space accessible through such modifications makes systematic exploration challenging, requiring computational approaches to accelerate their rational design and screening. Despite this potential, the limited availability of suitable computational models has left most ncAAs inaccessible to molecular dynamics simulations.

Here, we address this challenge by extending the Martini 3 coarse-grained force field, and its soon-to-be-released high resolution protein model, to incorporate over 250 non-canonical amino acids. We validate the extended model against available experimental and atomistic reference data and demonstrate its ability to capture structure–property relationships relevant to peptide membrane permeability and solubility. Our results substantially expand the chemical space accessible to Martini, enabling the study and rational engineering of chemically diverse peptides and proteins.